Dual-Branch Hypergraph Convolutional Network Learning Spectral–Spatial–Semantic Features for Hyperspectral Image Classification

Shuran Jing, Jinghua Li, Yijie Ding, Dehui Kong, and Baocai Yin

***Abstract*—Hyperspectral image (HSI) classification, which aims to assign pixel-level categories for given HSI data, has achieved remarkable success with deep learning architectures. However, such approaches typically require large amounts of annotated data, which are often scarce in practical applications. To address the challenge of limited annotated samples, this article proposes a novel framework that leverages both semantic and structural prior knowledge to extract more expressive HSI features. This article proposes a dual-branch hypergraph convolutional network (DB-HGCN) for comprehensive spectral–spatial–semantic feature extraction, which employs multihop constrained superpixel-level hypergraphs to effectively model the complex high-order correlations inherent in HSI data. The proposed architecture consists of two complementary branches: 1) a semantic-enhanced hypergraph convolutional network (SSeHGCN) that incorporates category-specific semantic knowledge through a vision–language model to enhance spatial representations and 2) a spatial–spectral enhanced hypergraph convolutional network (SSpHGCN) that captures both intra-superpixel visual features and their interrelationships via a novel spatial–spectral feature and relation fusion module (SSRFM). Extensive experiments on four benchmark datasets demonstrate the superiority of the proposed approach, with DB-HGCN achieving state-of-the-art overall classification accuracy (OA) using only five labeled samples per class. This significant performance gain highlights the effectiveness of our method in addressing the data scarcity challenge in HSI classification.**

***Index Terms*—Contrastive language–image pretraining (CLIP), hypergraph convolutional network (HGCN), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL remote sensing plays a vital role in geospatial applications by supporting land cover classification, environmental monitoring, and urban analysis [1], [2], [3], [4]. Hyperspectral images (HSIs), as 3-D data cubes with hundreds of bands, enable fine material discrimination but suffer from high dimensionality and complex class distributions. These challenges, combined with the scarcity of labeled samples in practical scenarios, make accurate HSI classification particularly difficult.

Early research relied on low-dimensional feature engineering and shallow models such as support vector machines (SVMs) [5] and random forests (RFs) [6], which struggle to capture the complex nonlinear spatial–spectral dependencies of HSI data. With the advent of deep learning [7], [8], convolution-based approaches have significantly improved HSI classification by learning joint spectral–spatial representations using 1-D [9], 2-D [10], and 3-D CNNs [11], [12], [13], [14]. Subsequent works have further enhanced these architectures by incorporating residual connections [15], hybrid convolutional branches [16], attention mechanisms [17], [18], and their combination [19].

More recently, Transformer-based frameworks [20], [21], [22], [23] have emerged as promising alternatives, offering stronger long-range dependency modeling and enhanced global context understanding. Notable advances include spectral sequence modeling, spatial–spectral feature tokenization, and group-aware hierarchical structures. Despite these developments, CNN-based methods still face challenges in receptive field limitations and generalization under scarce training data.

In contrast, graph convolutional networks (GCNs) [24] have demonstrated a distinct advantage by leveraging graph to incorporate additional structural information. GCN-based methods can be broadly classified into two categories: pixel level and superpixel level. Pixel-level methods [25], [26], [27] directly construct graphs where pixels serve as nodes. These methods define adjacency matrices by considering both spectral feature similarity and spatial proximity; however, they are prone to topological distortions, particularly in scenarios with limited sample sizes. In contrast, superpixel-level methods, which often use simple linear iterative clustering (SLIC) [28] to segment the original HSI image into superpixels, have demonstrated greater robustness in such situations. These approaches construct adjacency matrices based on the similarity of superpixel features. One of the pioneering works in this area is the multiscale dynamic GCN (MDGCN) proposed by Wan et al. [29]. Later works, such as the CNN-enhanced GCN (CEGCN) designed by Liu et al. [30], introduced a CNN-GCN dual-branch framework that can simultaneously extract both pixel- and superpixel-level features. Additionally, Dong et al. [31] proposed an attention fusion mechanism for CNN and GCN features in the weighted

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feature fusion of convolutional neural network and graph attention network (WFCG), improving the fusion effects. The multiscale spectral–spatial graph attention network (MSSGAT) [32] utilizes multiscale spectral–spatial neighborhood graphs to boost the classification accuracy. Zhou et al. [33] introduced the attention multihop graph and multiscale convolutional fusion network (AMGCFN), a multiscale convolution and multihop graph convolution fusion framework, which effectively extracts multiscale contextual features and significantly enhances the classification accuracy of complex land covers. While GCN offers substantial advantages in spatial modeling, its performance is limited by the binary adjacency matrix, which hinders its ability to adequately capture high-order associations typically found in HSI data.

Hypergraph theory [34] offers a solution to the limitations of CNNs and GCNs by allowing hyperedges to connect multiple nodes, enabling the explicit modeling of high-order associations and providing a richer representation of spatial–spectral relationships. Hypergraph-based HSI classification remains limited. Similar to GCN-based HSI classification, existing approaches can be divided into pixel- and superpixel-level methods. Pixel-level methods, such as the feature fusion hypergraph convolutional network (F2HNN) [35] and structure-preserved hypergraph convolution network (SPHGCN) [36], map pixels to hypergraph vertices and aggregate contextual information using hypergraph convolution. However, pixel-level methods tend to generate many hyperedges, which can increase the risk of noisy connections and overfitting, particularly in few-shot learning (FSL) scenarios. Superpixel-level methods, like those proposed by Xu et al. [37] and Wang et al. [38], address this by using SLIC segmentation and integrating CNN-based feature extraction. These methods perform well with abundant data but struggle in FSL settings due to noisy hyperedge connections.

To address HSI classification under limited sample conditions, a growing body of research has emerged for FSL directions. Many methods based on metric learning are applied in FSL scenarios and aim to learn a discriminative feature space by modeling relationships between samples. These include methods such as prototypical networks [39] and relation networks [40]. Contrastive learning is commonly employed to boost the few-shot classification performance of the aforementioned methods, thanks to its strong representation learning capacity [41], [42], [43]. Another important category is cross-domain FSL (CFSL), where knowledge from pretrained models is transferred to FSL tasks, helping mitigate domain shift and improve classification performance [44], [45], [46], [47]. Additionally, graph-based methods such as label propagation on graphs [48], [49] are also noteworthy in low-data scenarios. Different from the above HSI FSL paradigm, Li et al. [50] proposed a lightweight self-pooling Transformer (SPFormer), which tackles the FSL problem by reducing the number of model parameters, achieving excellent performance. Wang et al. [51] proposed the hypergraph convolutional network with multiple hyperedges fusion (HGCN-MHF), which introduced distance constraints between nodes, and has shown promising results in limited data scenarios. Inspired by this work, this article models HSI data with few samples via HGCN to learn its implicit structure knowledge. In addition, the aforementioned methods primarily focus on extracting spatial–spectral features from HSI visual data [52] and seldom incorporate prior semantic knowledge.

Semantic concepts (e.g., "forest" and "building") are inherently linked to the function or form of land covers and provide stable category descriptions across spectral variations [53], making them an ideal prior knowledge. Leveraging these concepts to enhance visual representation is a promising approach. Recently, the widely studied contrastive language–image pretraining (CLIP) model [54] has shown remarkable cross-modal alignment capabilities. By constructing a shared embedding space and utilizing large-scale image–text pairs for contrastive learning, CLIP establishes cross-modal alignment between visual features and textual descriptions, making it a powerful tool for zero-shot/few-shot classification tasks. This article introduces the CLIP model to improve visual feature representation in situations with limited samples by exploiting its implicit semantic knowledge.

As is known to us, semantic labels typically correspond to continuous spatial regions, such as the contiguous distribution of farmland. Hypergraph connects multiple nodes through hyperedges and is capable of capturing continuous semantics through multinode connections. Therefore, this article proposes a dual-branch hypergraph convolutional network (DB-HGCN), which attempts to leverage both structural and semantic knowledge to enhance feature representation, aiming to solve the problem of insufficient HSI data. The dual-branch framework separately supports semantic knowledge focusing and the retention of detailed spectral–spatial features. The main contributions of our work are as follows.

1) To address the challenge of limited annotated samples in HSI classification, we propose a dual-branch framework that jointly leverages semantic and structural priors. The proposed DB-HGCN enhances feature representation by incorporating implicit category-level semantics from CLIP and high-order structural correlations via multihop constrained hypergraph modeling.
2) To address the difficulty of modeling complex spatial–spectral dependencies under limited samples, we design a spatial–spectral feature and relation fusion module (SSRFM) that integrates both the internal visual features of superpixels and their inter-superpixel relationships, therefore constructing a more expressive spatial–spectral representation.
3) This study proposes a novel paradigm that combines semantic understanding with spatial–spectral analysis, enabling more expressive feature extraction under few-shot conditions. Experiments on four benchmark datasets demonstrate that the proposed model achieves superior performance in few-shot scenarios.

II. Method

For an HSI, we represent it as $X_{\text{hsi}} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the spatial height, width, and number of spectral bands of the HSI 3-D data cube, respectively. The one-hot encoded ground-truth label of each pixel is represented by $\tilde{Y} \in \mathbb{N}^{\text{HW}\times N_{\text{class}}}$, where N_{class} is the number of possible

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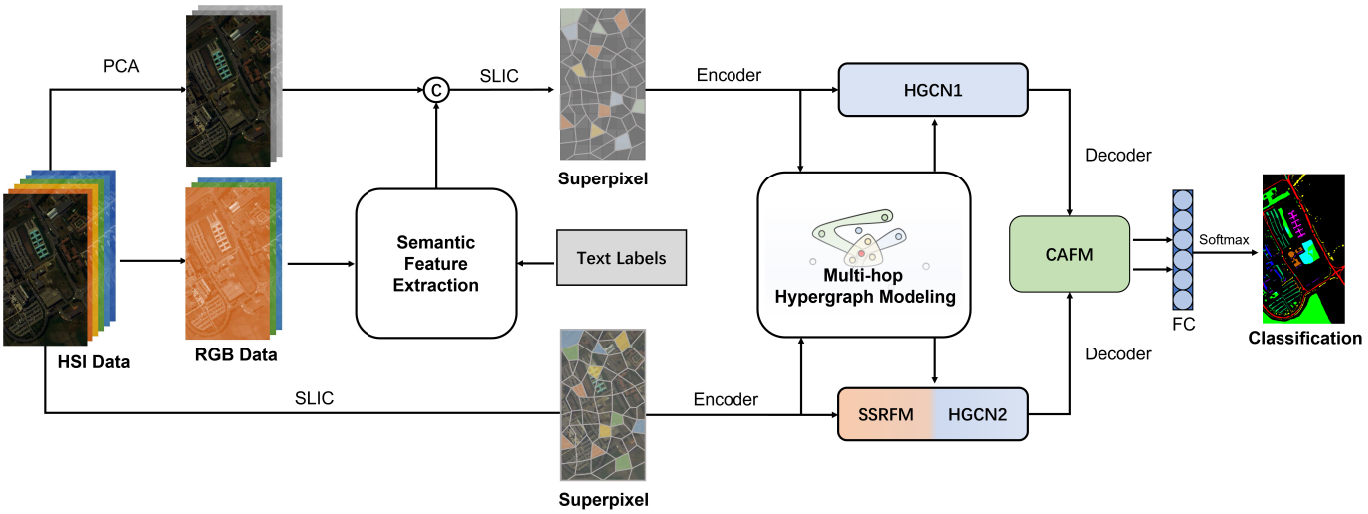


Fig. 1. Proposed DB-HGCN framework.

classes. The objective of the DB-HGCN is to learn an effective HSI feature representation to predict the class of each pixel as accurately as possible.

This article proposes a DB-HGCN. As shown in Fig. 1, the top branch is called semantic-enhanced hypergraph convolutional network (SSeHGCN) and the bottom one is called spatial–spectral enhanced hypergraph convolutional network (SSpHGCN). The SSeHGCN branch focuses on integrating semantic knowledge, so as to learn HSI spatial–semantic representation, while the SSpHGCN branch focuses to learn structure knowledge and enhances HSI spatial–spectral feature modeling. Specifically, for the input HSI data, the SSeHGCN branch extracts semantic features using CLIP and combines them with principal component analysis (PCA), feature reducing the dimensionality of the original spectral data. The SSpHGCN branch directly utilizes the raw HSI spectral information, thus preserving all the spectral details. Both branches use a multihop hypergraph convolution to extract features. Finally, the heterogeneous features extracted from the dual branches are integrated using cross-attention weighting and classified via softmax. In the following, we will elaborate on the design details of DB-HGCN.

A. HSI Spatial–Semantic Feature Extraction

Current HSI data analysis methods predominantly focus on spatial–spectral visual information while often neglecting the incorporation of semantic features. The cross-modal alignment capabilities of CLIP present a valuable opportunity to leverage semantic information in a more comprehensive manner. However, there are two key challenges in directly applying the CLIP model. First, the pretrained visual encoder of CLIP is designed for RGB images, whereas HSI involves a significantly different spectral composition. Second, pixel-level classification demands fine-grained semantic alignment, and the multiscale ground objects in HSI conflict with the fixed receptive field of CLIP [55], making direct extraction of fixed-resolution features through CLIP insufficient to meet these requirements. To address these challenges, this article proposes a method for generating semantic features for HSI using CLIP, as illustrated in Fig. 2.

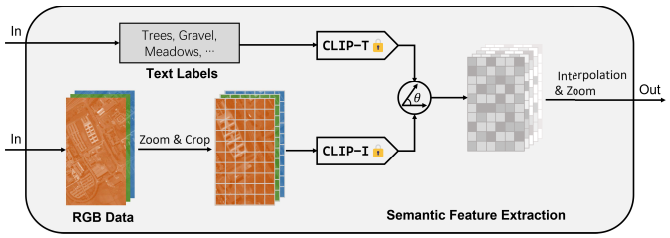


Fig. 2. Schematic of semantic feature extraction via CLIP.

In order to address the spectral dimension mismatch, we extract an equivalent RGB image $X_{\text{rgb}} \in \mathbb{R}^{H\times W\times 3}$ from the HSI cube using bands at 655 nm (red), 553 nm (green), and 451 nm (blue), enabling compatibility with the CLIP visual encoder. To enhance the recognition of both large-scale ground objects and small-scale targets, we apply a multiresolution strategy. Specifically, X_{rgb} is upsampled via cubic spline interpolation to multiple resolutions $\{Z_i\}_{i=1}^{L}$ (e.g., $2\times$ and $4\times$), producing $X_{\text{rgb}}^{Z_i} \in \mathbb{R}^{Z_iH\times Z_iW\times 3}$. For each resolution, we adopt a sliding window to partition the image into overlapping patches of size $K_c \times K_c$ and extract visual features using the CLIP visual encoder. Meanwhile, category labels are encoded into semantic embeddings via the CLIP text encoder. The semantic confidence map is computed by cosine similarity between visual features and class embeddings. To restore pixel-level resolution, each $P_{Z_i}^k$ is upsampled using bilinear interpolation and averaged across overlapping patches to form the semantic map $P_{Z_i} \in \mathbb{R}^{Z_iH\times Z_iW\times N_{\text{class}}}$. Finally, all P_{Z_i} maps are downsampled to the original resolution and averaged to obtain the multiscale CLIP semantic feature

$$X_{\text{clip}} = \frac{1}{L}\sum_{i=1}^{L} \text{Zoom}\left(P_{Z_i}, \frac{1}{Z_i}\right) \in \mathbb{R}^{H\times W\times N_{\text{class}}} \tag{1}$$

where $\text{Zoom}\left(\cdot, \frac{1}{Z_i}\right)$ denotes cubic spline downsampling. The resulting X_{clip} provides pixel-wise semantic priors aligned with HSI classification targets.

The representation capability of pure CLIP semantic features is constrained by the pretrained model, which may introduce interclass confusion. Consequently, there is a neces-

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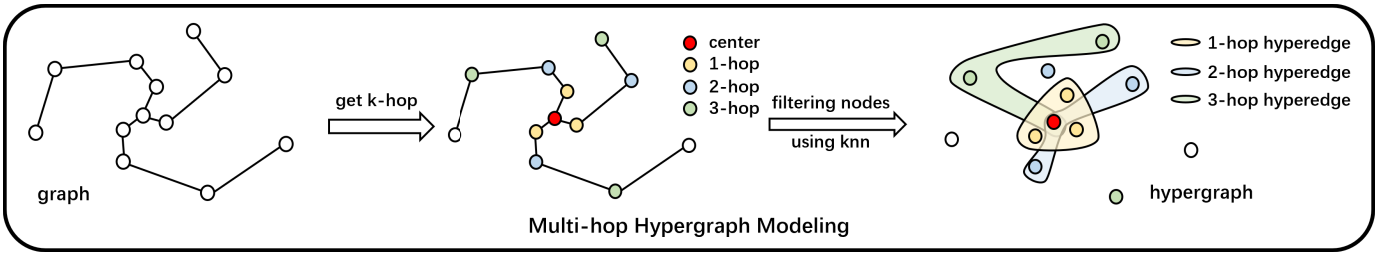


Fig. 3. Schematic of multihop hypergraph modeling.

sity to retain the original features for pixel-level guidance. To this end, we employ PCA to generate compact spatial–spectral features $X_{\text{pca}} \in \mathbb{R}^{H\times W\times A}$, where the selected PCA dimension $A \ll N_{\text{class}}$ ensures a balance between the original features and semantic information.

Finally, we concatenate the CLIP semantic features and PCA features along the channel dimension. This process enables the acquisition of pixel-level spatial–semantic enhanced features at the original spatial scale

$$X_{\text{sem}} = \left[X_{\text{pca}}; X_{\text{clip}}\right] \in \mathbb{R}^{H\times W\times (N_{\text{class}}+A)}. \tag{2}$$

B. Hypergraph Construction and Hypergraph Convolution Based on Multihop Constraints

Existing methods often construct hypergraphs based on superpixels and then extract features using graph convolution. These approaches struggle to capture nonneighbor similarities commonly found in land cover data (e.g., dispersed vegetation of the same class and cross-regional building clusters). However, hypergraph modeling tends to generate noisy connections, which can reduce the model's representational capacity. To address this issue, this article proposes structural regularization through multihop topological constraints. Specifically, the criteria of multihop constraints are derived from the connectivity patterns of the initial adjacency graph. A node is considered eligible for inclusion in a hyperedge only if it is reachable from the reference node within a predefined number of hops. This constraint regularizes the hypergraph topology by limiting redundant connections while preserving meaningful long-range correlations. The goal is to suppress noisy links and enhance semantic consistency across spatially separated but semantically related regions. The multihop hypergraph construction diagram is shown in Fig. 3.

The implementation of the proposed method involves three main stages. First, to maintain the homogeneity of the HSI local structure, we employ the classic SLIC algorithm for superpixel segmentation, treating the superpixels as nodes in the hypergraph. For the C-dimensional HSI cube $X \in \mathbb{R}^{H\times W\times C}$, we first flatten it into $\hat{X} \in \mathbb{R}^{\text{HW}\times C}$ and apply SLIC to generate N superpixels $\{S_j\}_{j=1}^{N}$. A pixel-superpixel mapping matrix $Q \in \{0,1\}^{\text{HW}\times N}$ is then constructed as

$$Q(i,j) = \begin{cases} 1, & \text{if } \hat{X}_i \in S_j \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

and the superpixel node feature matrix $V \in \mathbb{R}^{N\times C}$ can be computed by encoding $V = \bar{Q}^T \cdot \hat{X}$, where $\bar{Q}$ is the column-normalized mapping matrix Q.

Next, we construct the hypergraph based on the adjacency graph $G = (V, A)$ produced by superpixel segmentation, where $A \in \{0,1\}^{N\times N}$ represents the spatial adjacency matrix. By traversing each node in A with depth-first search, we can generate the k-hop neighborhood subgraph for each node

$$\mathcal{N}_{\text{hop}}^{k}(V_i) = \{V_i\} \cup \left\{V_j \middle| \min\left(\text{len}\left(V_i, V_j\right)\right) = k\right\} \tag{4}$$

where $\min(\text{len}(V_i, V_j))$ denotes the shortest path length between nodes V_i and V_j in A. We combine KNN hyperedge generation with the spatial topological constraints of the k-hop graph, selecting nodes based on feature similarity within the k-hop neighborhood of each central node to form the hyperedge set E

$$E = \bigcup_{i=1}^{N}\left(\bigcup_{k=2}^{\mathcal{K}}\left\{\text{KNN}\left(\mathcal{N}_{\text{hop}}^{k}(V_i)\right)\right\} \cup \left\{\mathcal{N}_{\text{hop}}^{1}(V_i)\right\}\right) \tag{5}$$

where $\text{KNN}(\cdot)$ denotes the KNN operation and $\mathcal{K}$ is the preselected number of hops. Given the specificity of the 1-hop neighborhood, it is necessary to treat all 1-hop nodes as a single hyperedge. This approach is taken to ensure the preservation of local detail features surrounding the central node. The resulting hypergraph is represented as $G_H = (V, E, W)$, where the node set V corresponds to N superpixel nodes, E is the hyperedge set corresponding to M hyperedges, and W is the weight matrix of the hyperedges. This hypergraph construction approach, in contrast to conventional global KNN computation, enables the extension of multiscale context through k-hop graph expansion. Furthermore, it allows for the restriction of the search range, reducing the likelihood of distance and noisy node misconnections.

Finally, the hypergraph incidence matrix $H \in \mathbb{R}^{N\times M}$ can be obtained through G_H

$$H(i,j) = \begin{cases} 1, & \text{if } V_i \in E_j \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

and the node degree matrix D_v and the hyperedge degree matrix D_e can be diagonalized as follows:

$$D_v(i,i) = \sum_{j}^{M} H(i,j)$$
$$D_e(j,j) = \sum_{i}^{N} H(i,j). \tag{7}$$

The hypergraph convolution part employs the general spatial convolution method (HGNN$^+$) [56] for information propagation and feature updating on the hypergraph structure. The

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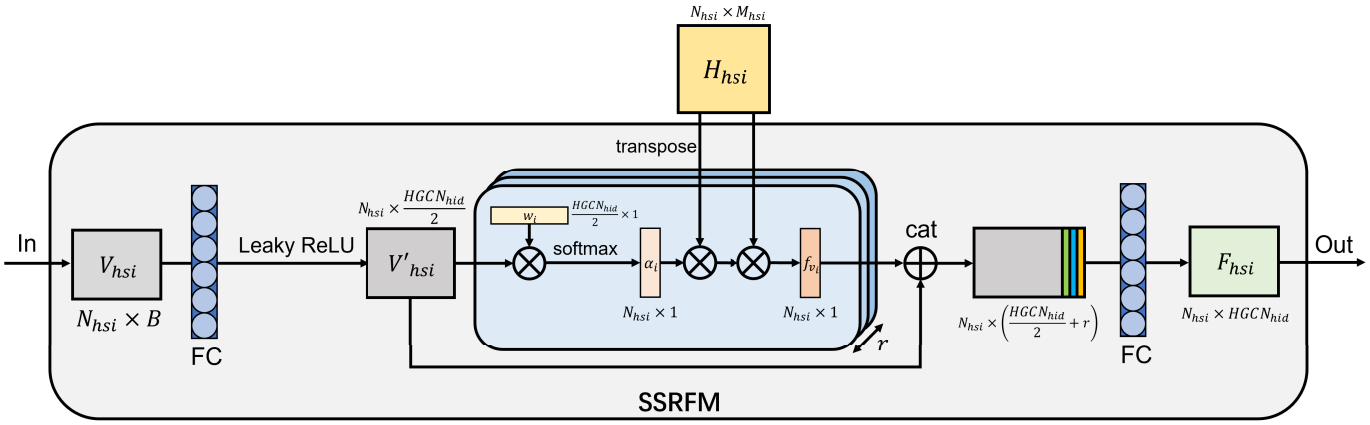


Fig. 4. Schematic of the SSRFM.

hypergraph convolution representation can be obtained based on the above hypergraph construction

$$F_{l+1} = \sigma\left(\mathrm{HGCN}\left(F_l\right)\right) = \sigma\left(D_v^{-1}\mathrm{HWD}_e^{-1}H^T F_l \Theta_l\right) \tag{8}$$

where Θ_l is the trainable weight matrix of the lth layer; F_{l+1} and F_l are the superpixel node features of the $l+1$th and lth layers in the hypergraph, respectively; and $\sigma(\cdot)$ denotes the LeakyRelu activation function.

C. Spatial–Spectral and Relation Feature Extracting

Building on the aforementioned multihop hypergraph construction method, we apply it to both the spatial–semantic features X_{sem} and the original HSI data X_{hsi}, resulting in the respective hypergraphs $G_H^{\mathrm{sem}} = (V_{\mathrm{sem}}, E_{\mathrm{sem}}, W_{\mathrm{sem}})$ and $G_H^{\mathrm{hsi}} = (V_{\mathrm{hsi}}, E_{\mathrm{hsi}}, W_{\mathrm{hsi}})$. However, the information modeled by the two branches proposed in this article is significantly different. Specifically, the SSeHGCN branch extracts heterogeneous features comprising class confidence descriptions and PCA features, while the SSpHGCN branch models all spectral information, which encompasses complex spectral heterogeneity and a substantial amount of noise. The distinct characteristics of the two superpixel node features, V_{sem} and V_{hsi}, necessitate the design of their respective dimension transformation modules for dimensionality reduction and noise reduction, thereby facilitating the subsequent hypergraph convolution feature extraction.

For the spatial–semantic features $V_{\mathrm{sem}} \in \mathbb{R}^{N_{\mathrm{sem}} \times (N_{\mathrm{class}} + A)}$ of the SSeHGCN branch, we use a fully connected layer for linear transformation to map the $(N_{\mathrm{class}} + A)$-dimensional heterogeneous features to a unified space

$$F_{\mathrm{sem}} = \mathrm{FC}\left(V_{\mathrm{sem}}\right) = W_1 V_{\mathrm{sem}} + B_1 \in \mathbb{R}^{N_{\mathrm{sem}} \times \mathrm{HGCN}_{\mathrm{hid}}} \tag{9}$$

where W_1 and B_1 represent the weights and biases of the fully connected layer FC(·), respectively; N_{sem} is the number of superpixel nodes in V_{sem}; and $\mathrm{HGCN}_{\mathrm{hid}}$ denotes the hidden dimension of the convolutional layer.

In the context of the spatial–spectral characteristics of the SSpHGCN branch, the lack of an effective hyperedge weighting strategy during hypergraph construction, designed to mitigate overfitting, leads to insufficient modeling of the interaction intensity between nodes. As a result, substantial spectral noise interference arises. To address this issue, we propose the SSRFM, as illustrated in Fig. 4. This module extracts spatial–spectral relational features through multiple independent weighting paths and concatenates them into unique feature vectors, which are then fused with the original features. The core idea behind this module is to use learnable weights to guide two-phase feature propagation and enhance class-discriminative features. This approach not only helps reduce dimensionality and noise but also aids the hypergraph convolution process in feature extraction, especially in scenarios with limited samples.

Given the input HSI superpixel feature matrix $V_{\mathrm{hsi}} \in \mathbb{R}^{N_{\mathrm{hsi}} \times B}$, feature encoding is first performed through linear transformation

$$V'_{\mathrm{hsi}} = \sigma\left(\mathrm{FC}\left(V_{\mathrm{hsi}}\right)\right) = \sigma\left(W_2 V_{\mathrm{hsi}} + B_2\right) \in \mathbb{R}^{N_{\mathrm{hsi}} \times \frac{\mathrm{HGCN}_{\mathrm{hid}}}{2}} \tag{10}$$

where W_2 and B_2 represent the weights and biases in the linear layer, respectively, and $\sigma(\cdot)$ denotes the LeakyRelu activation function.

The proposed approach constructs r independent weighting paths, each corresponding to a learnable weight. For each path, the node importance distribution is calculated through the learnable weight vector

$$\alpha_i = \mathrm{Softmax}\left(V'_{\mathrm{hsi}} w_i\right) \in \mathbb{R}^{N_{\mathrm{hsi}} \times 1} \tag{11}$$

where $w_i \in \mathbb{R}^{\frac{\mathrm{HGCN}_{\mathrm{hid}}}{2} \times 1}$ is the exclusive weight vector for the ith path. Two-phase weighted aggregation is then performed from node to hyperedge and back to node based on the hypergraph incidence matrix $H_{\mathrm{hsi}} \in \mathbb{R}^{N_{\mathrm{hsi}} \times M_{\mathrm{hsi}}}$

$$f_{e_i} = H_{\mathrm{hsi}}^T \alpha_i \in \mathbb{R}^{M_{\mathrm{hsi}} \times 1} \tag{12}$$

$$f_{v_i} = H_{\mathrm{hsi}} f_{e_i} \in \mathbb{R}^{N_{\mathrm{hsi}} \times 1} \tag{13}$$

where f_e and f_v represent the hyperedge feature weights and propagated node weights, respectively.

Finally, concatenate the outputs of each path with the original features and fuse through a linear transformation

$$F_{\mathrm{hsi}} = W_3 \left[V'_{\mathrm{hsi}} : \bigoplus_{i=1}^{r} f_{v_i} \right] + B_3 \in \mathbb{R}^{N_{\mathrm{hsi}} \times \mathrm{HGCN}_{\mathrm{hid}}} \tag{14}$$

where W_3 and B_3 are the weights and biases of the linear layer, respectively, and $\bigoplus$ denotes the concatenation operation.

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Through concatenation and linear transformation, the final fused features retain the spatial consistency of the original encoded features and add newly constructed relational information, enhancing the overall feature expression capabilities.

It is important to acknowledge that the SSRFM module cannot be applied to the SSeHGCN branch. This is due to the fact that the semantic features V_{sem} are a low-dimensional compact representation that is achieved through CLIP and PCA. The implementation of the SSRFM's multipath concatenation operation would result in the disruption of the semantic features' simplicity and the potential introduction of redundant noise.

HGCN is employed in both branches to extract key features, with each layer followed by batch normalization (BN) and activation to enhance generalization and training stability. The resulting encoded features are

$$F_{\text{sem}}^{\text{cov}}, F_{\text{hsi}}^{\text{cov}} = \sigma\left(\text{BN}\left(\text{HGCN}\left(\sigma\left(\text{BN}\left(\text{HGCN}\left(F_{\text{sem}}, F_{\text{hsi}}\right)\right)\right)\right)\right)\right). \tag{15}$$

The outputs are then projected back to pixel space using the corresponding pixel-to-superpixel mapping matrices Q_{sem} and Q_{hsi}, yielding $F_{\text{sem}}^{\text{out}}, F_{\text{hsi}}^{\text{out}} \in \mathbb{R}^{\text{HW}\times\text{HGCN}_{\text{hid}}}$.

D. Classification and Loss Function

To fuse the features from both branches, the cross-attention feature fusion module (CAFM) [33] is employed to compute weighted outputs for the semantic and HSI branches as $Y_{\text{sem}}^{\text{out}}$ and $Y_{\text{hsi}}^{\text{out}}$. These outputs are concatenated to form the fused feature

$$Y_{\text{cross}}^{\text{out}} = \left[Y_{\text{sem}}^{\text{out}}; Y_{\text{hsi}}^{\text{out}}\right] \in \mathbb{R}^{\text{HW}\times 2\ \text{HGCN}_{\text{hid}}}. \tag{16}$$

The final classification result is obtained through a fully connected layer followed by a softmax function

$$Y = \text{Softmax}\left(\text{FC}\left(Y_{\text{cross}}^{\text{out}}\right)\right). \tag{17}$$

In the network training for multiclassification tasks, the general cross-entropy loss is utilized to quantify the discrepancy between the category predictions and the true labels

$$\mathcal{L} = -\sum_{t=1}^{\text{HW}}\sum_{y=1}^{N_{\text{class}}} \tilde{Y}(t,y)\log Y(t,y) \tag{18}$$

where $\tilde{Y}(t,y) \in \{0,1\}$ represents the one-hot encoded ground-truth label for the tth pixel. If the tth pixel belongs to class y, then $\tilde{Y}(t,y) = 1$; otherwise, $\tilde{Y}(t,y) = 0$. $Y(t,y) \in [0,1]$ represents the predicted probability that the tth pixel belongs to class y.

However, due to the relative independence of feature extraction in the dual-branch structure, balancing the losses of both branches using a simple loss function becomes challenging. To address this issue, we propose incorporating independent losses for the two feature extraction streams as auxiliaries, in addition to the overall cross-entropy loss. The resulting loss function is expressed as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DB-HGCN}} + \mathcal{L}_{\text{SSeHGCN}} + \mathcal{L}_{\text{SSpHGCN}} \tag{19}$$

where $\mathcal{L}_{\text{DB-HGCN}}$ is the main cross-entropy loss for the final classification output from the DB-HGCN network and $\mathcal{L}_{\text{SSeHGCN}}$ and $\mathcal{L}_{\text{SSpHGCN}}$ are the auxiliary losses for the SSeHGCN and SSpHGCN branches, respectively.

III. Experiments and Analysis

To verify and evaluate the effectiveness of the DB-HGCN model, both conventional experiments and ablation studies were designed and analyzed on four public hyperspectral datasets.

A. Dataset Description

The four public datasets used are introduced as follows, with detailed class information provided in Table I.

1) University of Pavia: Acquired by ROSIS-03 over Pavia, Italy, in 2003, with a spatial resolution of 1.3 m. The image has dimensions of 610 × 340 pixels and 103 usable spectral bands. It contains 42 776 samples across nine land cover classes.

2) WHU-Hi-HongHu: Captured in Honghu, China, using a UAV-mounted Headwall Nano-Hyperspec sensor. The image size is 940 × 475 pixels, covering 270 bands (400–1000 nm) with 0.043-m spatial resolution. It includes 22 land cover classes.

3) Salinas: Collected by the AVIRIS sensor over the Salinas (SA) Valley, California. The image spans 512 × 217 pixels, with 204 usable bands and 3.7-m spatial resolution. It includes 54 129 labeled samples across 16 classes.

4) WHU-Hi-HanChuan: Acquired in Hanchuan, China, using a UAV-based Headwall Nano-Hyperspec sensor. The image size is 1217 × 303 pixels, with 274 bands (400–1000 nm) and 0.109-m spatial resolution. It contains 16 land cover classes, with notable shadowed regions due to low solar angle.

B. Experimental Settings

1) Training Setting: To maintain consistency with mainstream experimental settings, refer to [50], five samples were randomly selected from each category in each experimental dataset to construct the training set. The remaining samples were then used as the test set to evaluate classification accuracy. For the CLIP component, we implemented a multiscale enhancement procedure, employing resolutions twice and four times the original image size. The enhanced images were then cropped with a window size of 112 × 112 and a stride of 56 × 56. The PCA retention dimension for aggregation was set to 3. The hidden layer dimension HGCN_{hid} for the convolutional part was set to 64. The learning rates for the University of Pavia (UP), WHU-Hi-HongHu (HH), SA, and WHU-Hi-HanChuan (HC) datasets were set to 1e−3, 5e−4, 1e−4, and 5e−4, respectively. The total number of training epochs was set to 400, with the Adam optimizer employed for gradient iteration during each epoch.

2) Comparison Methods: We selected several mainstream and state-of-the-art methods to compare with DB-HGCN in the hyperspectral FSL task to verify its effectiveness.

1) *MambaHSI [57]:* A model based on the Mamba architecture that performs image-level spatial–spectral modeling.
2) *MLPA [58]:* A CFSL method that utilizes multilevel prototype alignment.
3) *SSMAE [59]:* A dual-phase framework combining a self-supervised masked autoencoder and hybrid episodic training.

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TABLE I
CATEGORY INFORMATION OF FOUR DATASETS

#	Colors	WHU-Hi-HongHu		University of Pavia		Salinas		WHU-Hi-HanChuan	
		Label	Samples	Label	Samples	Label	Samples	Label	Samples
1		Red roof	14041	Asphalt	6631	Brocoll green weeds 1	2009	Strawberry field	44735
2		Road	3512	Meadows	18649	Brocoll green weeds 2	3726	Cowpea field	22753
3		Bare soil	21821	Gravel	2099	Fallow	1976	Soybean field	10287
4		Cotton field	163285	Trees	3064	Fallow rough plow	1394	Sorghum field	5353
5		Cotton firewood field	6218	Painted metal sheets	1345	Fallow smooth	2678	Water spinach field	1200
6		Rape field	44557	Bare Soil	5029	Stubble	3959	Watermelon field	4533
7		Chinese cabbage field	24103	Bitumen	1330	Celery	3579	Greens field	5903
8		Pakchoi field	4054	Self-Blocking Bricks	3682	Grapes untrained	11271	Trees	17978
9		Cabbage field	10819	Shadows	947	Soil vinyard develop	6203	Grass	9469
10		Tuber mustard field	12394			Corn senesced green weeds	3278	Red roof	10516
11		Brassica parachinensis field	11015			Lettuce romaine 4wk	1068	Gray roof	16911
12		Brassica chinensis field	8954			Lettuce romaine 5wk	1927	Plastic	3679
13		Small Brassica chinensis field	22507			Lettuce romaine 6wk	916	Bare soil	9116
14		Lactuca sativa field	7356			Lettuce romaine 7wk	1070	Road	18560
15		Celtuce field	1002			Vinyard untrained	7268	Bright object	1136
16		Film covered lettuce field	7262			Vinyard vertical trellis	1807	Water	75401
17		Romaine lettuce field	3010						
18		Carrot field	3217						
19		White radish field	8712						
20		Garlic sprout field	3486						
21		Broad bean field	1328						
22		Tree	4040						
		Total	**386693**	**Total**	**42776**	**Total**	**54129**	**Total**	**257530**

4) *SSEFN [52]:* A model that incorporates spatial–spectral enhancement and decision fusion for limited sample settings.
5) *SPFormer [50]:* A lightweight Transformer method for few-shot scenarios, incorporating self-pooling and parameter-free channel mixing.
6) *AMGCFN [33]:* A model combining multihop graph attention and multiscale convolutional fusion.
7) *CEGCN [30]:* A hybrid model integrating CNN and GCN to extract complementary pixel- and superpixel-level features.
8) *GAHT [23]:* A group-aware hierarchical Transformer model for local spatial–spectral context modeling.
9) *WSGraphCL [60]:* A weak–strong graph contrastive learning model based on unsupervised graph-based pretraining.
10) *Gia-CFSL [46]:* A CFSL method combining graph information aggregation and domain alignment.

The parameters of all comparison methods were set according to the original articles, while the experimental sample settings were consistent with those in this study.

3) Evaluation Metrics: Three widely employed metrics are selected to evaluate the classification accuracy: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kpp). The presented results of ten independent experiments are averaged.

4) Running Platforms: All the experiments are implemented by PyTorch 2.3.1 with CUDA11.8 in Python 3.10.16 environment, and the hardware used for training is an i9-13900K CPU and a NVIDIA GeForce RTX4090 GPU with 24-GB RAM.

C. Hyperparameter Analysis

The selection of hyperparameters exerts a substantial influence on model performance. The present study focuses on a discussion of four types of hyperparameters: SLIC segmentation scale (λ), KNN neighborhood number (K), multihop constraint scale (hop), and the number of weights of the SSRFM (r).

1) Analysis of λ: The SLIC superpixel segmentation scale, λ, determines the granularity of the superpixels, directly influencing the local homogeneity of hypergraph nodes and the global association range, which forms the foundation for hypergraph construction. To identify the optimal segmentation scale, we varied λ within the range $[100, 200, 300, 400, 500]$ and recorded the OA across the four datasets, as shown in Fig. 5(a). The experimental results revealed that the highest OA for the UP, HH, SA, and HC datasets was achieved when λ was set to 400, 100, 500, and 200, respectively.

2) Analysis of K: The number of KNN neighbors, K, combined with multihop topological constraints, strikes a balance between capturing high-order associations and suppressing noise. To determine the optimal value of K, we conducted experiments with K values ranging from 3 to 11 across the four datasets, with the resulting OA shown in Fig. 5(b). As observed, the optimal choice of K for the UP, HH, SA, and HC datasets was 5. Notably, since the KNN was constrained by the multihop neighborhood, the OA performance across the four datasets showed minimal fluctuation with changes in K.

3) Analysis of Hop: The multihop constraint scale, hop, determines the topological depth of hypergraph connectivity, affecting the model's ability to capture high-order semantic associations while suppressing noisy links. As shown in Fig. 5(c), we evaluated the OA across the four datasets using different hop sizes ranging from 1 to 5. The optimal performance was achieved with a four-hop setting on the UP dataset and a three-hop setting on the HH, SA, and HC datasets. These findings suggest that a limited hop scale may restrict the expressive power of the hypergraph, failing to capture long-range correlations, whereas an overly large hop scale may introduce redundant or irrelevant connections, leading to performance degradation. Therefore, a moderate hop value is crucial to maintain a balance between structural richness and noise suppression in hypergraph learning.

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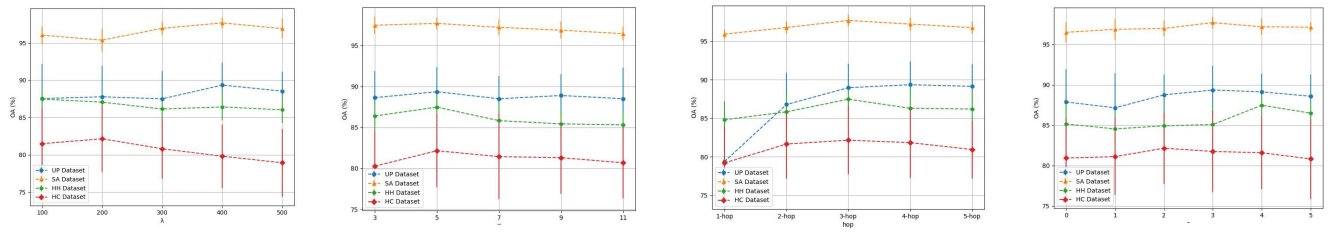


Fig. 5. Evolution of overall accuracy (OA) as a function of (a) superpixel segmentation scale λ, (b) KNN neighborhood number K, (c) multihop constraint scale hop, and (d) number of weights r.

TABLE II
CLASSIFICATION RESULTS ON THE UP DATASET

Class	MambaHSI [57]	MLPA [58]	SSMAE [59]	SSEFN [52]	SPFormer [50]	AMGCFN [33]	CEGCN [30]	GAHT [23]	WSGraphCL [60]	Gia-CFSL [46]	DB-HGCN(ours)
1	75.77	79.66	82.19	**97.47**	86.41	85.03	80.24	81.91	78.17	75.16	81.62
2	76.99	84.45	85.84	**96.35**	74.01	74.99	78.57	68.86	73.71	89.68	87.56
3	68.17	65.67	83.88	61.11	80.77	93.63	90.98	56.99	86.24	72.83	**98.81**
4	77.92	92.53	93.79	73.68	91.73	87.83	79.48	**96.49**	88.67	90.65	85.25
5	99.53	99.24	98.09	99.65	**99.97**	99.91	99.81	99.90	94.74	99.62	99.47
6	90.62	78.64	96.86	68.62	93.09	97.71	94.46	73.95	84.13	72.28	**98.55**
7	83.91	78.01	87.05	73.24	99.48	98.84	95.98	86.39	89.05	77.63	**99.54**
8	80.95	62.62	81.20	74.44	89.97	90.80	86.88	78.83	90.34	69.97	**93.85**
9	97.54	98.56	96.37	91.60	97.06	91.29	81.12	**99.57**	84.35	97.80	76.00
OA(%)	79.75 ± 4.71	81.38 ± 3.10	87.30 ± 3.12	82.02 ± 6.20	83.26 ± 3.98	84.29 ± 6.70	83.35 ± 2.63	75.92 ± 4.03	80.12 ± 4.01	82.94 ± 1.99	$\mathbf{89.35 \pm 2.98}$
AA(%)	83.49 ± 2.72	82.15 ± 1.66	89.47 ± 0.92	81.80 ± 3.21	90.28 ± 2.53	91.11 ± 3.20	87.50 ± 1.62	82.54 ± 1.97	85.49 ± 2.48	82.85 ± 1.93	$\mathbf{91.18 \pm 1.74}$
Kpp($\times$100)	82.00 ± 5.48	75.97 ± 3.64	83.69 ± 3.79	77.36 ± 7.03	79.02 ± 4.62	80.37 ± 7.93	78.75 ± 3.15	69.79 ± 4.55	75.02 ± 4.58	76.97 ± 2.42	$\mathbf{86.27 \pm 3.70}$

TABLE III
CLASSIFICATION RESULTS ON THE HH DATASET

Class	MambaHSI [57]	SSMAE [59]	SSEFN [52]	SPFormer [50]	AMGCFN [33]	CEGCN [30]	GAHT [23]	WSGraphCL [60]	DB-HGCN(ours)
1	81.31	91.03	**97.83**	84.38	85.82	80.28	83.50	82.33	88.82
2	70.15	82.55	62.37	**92.74**	63.82	84.12	82.49	78.13	87.89
3	81.45	87.41	**89.94**	81.83	84.00	87.35	73.32	83.80	89.19
4	88.19	96.99	**99.29**	90.26	94.15	86.77	85.66	88.01	91.47
5	83.31	95.45	30.69	93.73	95.42	98.25	79.79	89.34	**98.74**
6	81.70	92.00	**92.90**	83.36	91.09	90.66	86.62	85.53	91.49
7	55.36	77.51	**88.31**	72.05	51.93	63.53	48.75	59.85	66.94
8	66.91	81.43	52.70	81.61	84.56	48.97	32.47	78.74	**95.18**
9	82.50	91.26	**97.62**	92.73	88.86	92.69	91.98	84.70	92.55
10	58.19	**91.94**	88.25	86.86	78.75	72.10	74.28	76.94	82.95
11	56.12	67.87	72.60	69.44	56.61	56.78	49.50	**75.15**	72.67
12	58.20	**81.01**	46.88	67.38	71.76	74.35	60.69	61.54	80.05
13	67.65	66.49	**76.52**	59.81	58.85	65.20	56.61	67.07	71.78
14	70.85	**96.16**	74.97	67.86	66.30	71.39	75.98	74.56	85.04
15	95.55	94.08	41.59	96.18	94.66	96.09	92.17	84.02	**97.12**
16	76.21	98.02	91.85	86.50	71.40	84.40	79.62	94.62	**98.98**
17	71.21	78.17	64.57	67.16	85.42	87.01	73.39	66.10	**85.89**
18	88.77	94.91	58.17	84.29	96.49	95.68	76.48	82.17	**96.84**
19	71.91	**94.90**	76.85	84.62	65.87	86.33	77.85	77.12	76.44
20	90.97	54.55	64.32	90.06	93.00	88.65	77.83	81.86	**95.20**
21	93.65	92.97	28.41	91.74	99.76	95.90	77.69	99.28	**100.00**
22	86.03	88.20	41.57	79.65	**99.13**	97.59	83.53	85.40	96.66
OA(%)	79.37 ± 2.22	86.17 ± 4.23	83.08 ± 3.08	83.75 ± 1.06	84.70 ± 1.74	82.55 ± 2.70	77.78 ± 4.02	81.83 ± 2.02	$\mathbf{87.47 \pm 1.63}$
AA(%)	76.19 ± 2.72	86.13 ± 3.29	69.92 ± 2.38	82.01 ± 0.55	80.80 ± 1.61	82.00 ± 1.97	73.65 ± 2.17	79.83 ± 2.97	$\mathbf{88.27 \pm 1.03}$
Kpp($\times$100)	73.24 ± 3.08	83.67 ± 4.52	79.20 ± 3.48	79.95 ± 1.19	80.13 ± 2.07	78.63 ± 3.05	72.90 ± 4.52	77.71 ± 2.23	$\mathbf{84.54 \pm 1.88}$

4) Analysis of r: The number of weights, r, in the SSRFM module plays a crucial role in determining the diversity of multipath feature aggregation, which influences the ability to model complex spectral relationships. To assess the impact of r, we set it to $[0, 1, 2, 3, 4, 5]$ and conducted experiments on the four datasets, as shown in Fig. 5(d). When $r = 0$, the features pass directly through two linear layers without engaging in additional weight learning stages. The OA results were achieved at $r = 3$ for both UP and SA datasets, while HH required $r = 4$ and HC performed best at $r = 2$. Interestingly, when $r = 1$, the OA on the UP and HH datasets was lower than when $r = 0$, suggesting that a single-path weight could introduce bias in feature representation. For r values exceeding their respective optima (3 for UP/SA, 4 for HH, and 2 for HC), all datasets showed performance degradation, indicating that redundant weights might blur the decision boundaries.

D. Classification Results

To demonstrate the classification effectiveness of DB-HGCN in limited sample scenarios, we compared its performance with ten classic models across the UP, HH, SA, and HC datasets, using five training samples per class. The experimental results are presented in Tables II–V. The analysis of the results is as follows.

1) UP Dataset: On the UP dataset, DB-HGCN achieves the best performance with an OA of 89.35%, improving by 2.05% to 13.43% over other models. The AA and Kpp

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TABLE IV

CLASSIFICATION RESULTS ON THE SA DATASET

Class	MambaHSI [57]	MLPA [58]	SSMAE [59]	SSEFN [52]	SPFormer [50]	AMGCFN [33]	CEGCN [30]	GAHT [23]	WSGraphCL [60]	Gia-CFSL [46]	DB-HGCN(ours)
1	96.41	99.07	99.70	**100.00**	99.94	98.36	99.49	98.36	97.44	97.68	**100.00**
2	98.93	99.12	98.67	99.25	**100.00**	98.42	**100.00**	96.42	97.89	98.49	99.99
3	94.28	88.57	99.05	97.49	91.55	97.90	**99.87**	93.13	86.68	89.66	98.51
4	98.87	99.41	94.23	96.98	99.96	97.02	99.75	99.26	93.77	98.71	98.95
5	97.38	91.06	90.71	96.45	97.44	91.04	96.36	91.00	90.44	88.38	**97.64**
6	99.37	98.83	94.48	**100.00**	99.45	95.61	99.48	99.46	98.08	99.02	98.99
7	98.42	99.45	99.22	99.82	99.97	98.41	99.87	99.26	98.07	98.59	**100.00**
8	76.26	75.45	69.77	77.06	76.97	88.70	74.83	62.38	75.52	78.18	**96.15**
9	99.25	99.01	99.23	99.40	99.71	99.95	99.96	98.34	99.75	98.54	**100.00**
10	90.37	84.96	94.35	95.17	92.51	86.85	91.90	89.16	96.23	79.98	**96.25**
11	95.86	97.81	98.61	94.16	**99.96**	96.45	98.10	97.81	93.47	97.86	97.13
12	99.46	98.05	97.25	98.98	98.91	96.56	**99.94**	95.68	99.28	99.06	97.00
13	99.09	98.65	99.19	95.04	98.74	93.39	**99.94**	98.80	96.91	96.89	98.01
14	99.18	98.68	**99.59**	93.81	98.53	96.37	95.91	97.52	89.14	97.43	94.39
15	83.57	74.47	84.02	63.73	81.64	84.54	85.62	73.84	60.20	73.55	**94.73**
16	96.61	89.80	97.23	99.96	99.30	99.96	98.05	92.17	97.38	89.82	**100.00**
OA(%)	91.19 ± 1.82	88.83 ± 2.29	90.68 ± 1.04	88.31 ± 2.03	91.65 ± 1.91	92.21 ± 1.28	91.88 ± 2.41	86.15 ± 2.45	90.43 ± 2.78	88.57 ± 1.24	$\mathbf{97.69 \pm 0.72}$
AA(%)	95.21 ± 0.87	93.27 ± 1.57	94.70 ± 0.38	94.21 ± 0.82	95.91 ± 0.91	94.97 ± 2.28	96.19 ± 0.92	92.66 ± 1.41	91.89 ± 2.41	92.62 ± 0.91	$\mathbf{97.99 \pm 0.56}$
Kpp(×100)	94.87 ± 1.77	87.60 ± 2.51	88.56 ± 1.12	86.99 ± 2.23	90.73 ± 2.09	91.73 ± 1.43	90.99 ± 2.66	84.64 ± 2.72	89.39 ± 3.04	85.79 ± 1.39	$\mathbf{97.42 \pm 0.80}$

TABLE V

CLASSIFICATION RESULTS ON THE WHU-HI-HANCHUAN DATASET

Class	MambaHSI [57]	MLPA [58]	SSMAE [59]	SSEFN [52]	SPFormer [50]	AMGCFN [33]	CEGCN [30]	GAHT [23]	WSGraphCL [60]	Gia-CFSL [46]	DB-HGCN(ours)
1	70.67	76.33	66.24	**93.96**	70.73	88.34	87.83	77.64	73.91	74.05	83.43
2	45.16	31.83	82.71	**92.68**	62.48	55.25	41.20	64.20	61.66	37.53	69.45
3	74.07	62.83	78.90	69.93	82.94	73.26	88.95	60.75	75.41	65.74	**90.09**
4	94.26	90.43	**98.83**	76.28	94.89	94.32	98.40	91.75	96.03	89.47	94.46
5	96.34	80.16	94.50	26.46	97.94	96.09	**98.62**	87.50	91.67	79.11	98.35
6	42.81	30.53	68.44	38.89	65.61	**88.14**	67.65	28.24	62.29	34.96	83.43
7	75.45	84.50	90.81	51.72	91.33	**93.61**	91.52	78.61	86.66	74.20	91.61
8	36.88	38.07	60.59	82.66	63.11	70.46	61.13	47.85	58.96	38.37	**83.53**
9	50.77	30.54	72.07	68.83	**78.67**	59.08	40.49	29.96	51.43	32.08	68.26
10	72.47	77.73	95.27	83.54	**98.23**	74.57	82.10	82.94	73.10	81.00	87.64
11	63.45	63.25	**89.78**	53.54	89.35	88.63	87.94	62.50	75.28	59.34	83.01
12	68.12	47.38	80.23	43.47	71.85	92.20	76.35	61.97	77.69	58.64	**92.94**
13	53.21	36.00	63.96	48.29	47.54	63.00	44.82	37.01	53.38	35.94	**67.87**
14	51.29	64.24	82.90	**73.71**	41.43	58.75	50.98	51.99	50.27	65.50	69.53
15	75.68	70.03	83.55	49.47	**84.09**	61.18	73.33	82.54	83.93	70.93	78.31
16	90.96	88.00	94.98	98.90	**99.73**	89.04	90.72	91.33	98.15	89.63	87.07
OA(%)	69.20 ± 3.64	67.48 ± 3.16	80.36 ± 4.77	75.80 ± 3.30	79.24 ± 1.09	79.33 ± 2.62	76.36 ± 3.95	71.18 ± 4.16	74.21 ± 3.78	67.17 ± 3.48	$\mathbf{82.15 \pm 4.44}$
AA(%)	66.35 ± 2.96	60.50 ± 2.20	81.49 ± 5.07	65.77 ± 2.13	77.50 ± 0.74	77.87 ± 1.57	73.88 ± 2.54	64.80 ± 2.75	73.11 ± 6.10	61.66 ± 3.17	$\mathbf{83.06 \pm 2.35}$
Kpp(×100)	64.20 ± 5.18	62.94 ± 3.06	79.06 ± 4.15	72.16 ± 3.59	75.96 ± 1.25	76.09 ± 2.86	72.71 ± 4.37	66.66 ± 4.48	70.29 ± 4.21	62.21 ± 3.71	$\mathbf{79.50 \pm 4.95}$

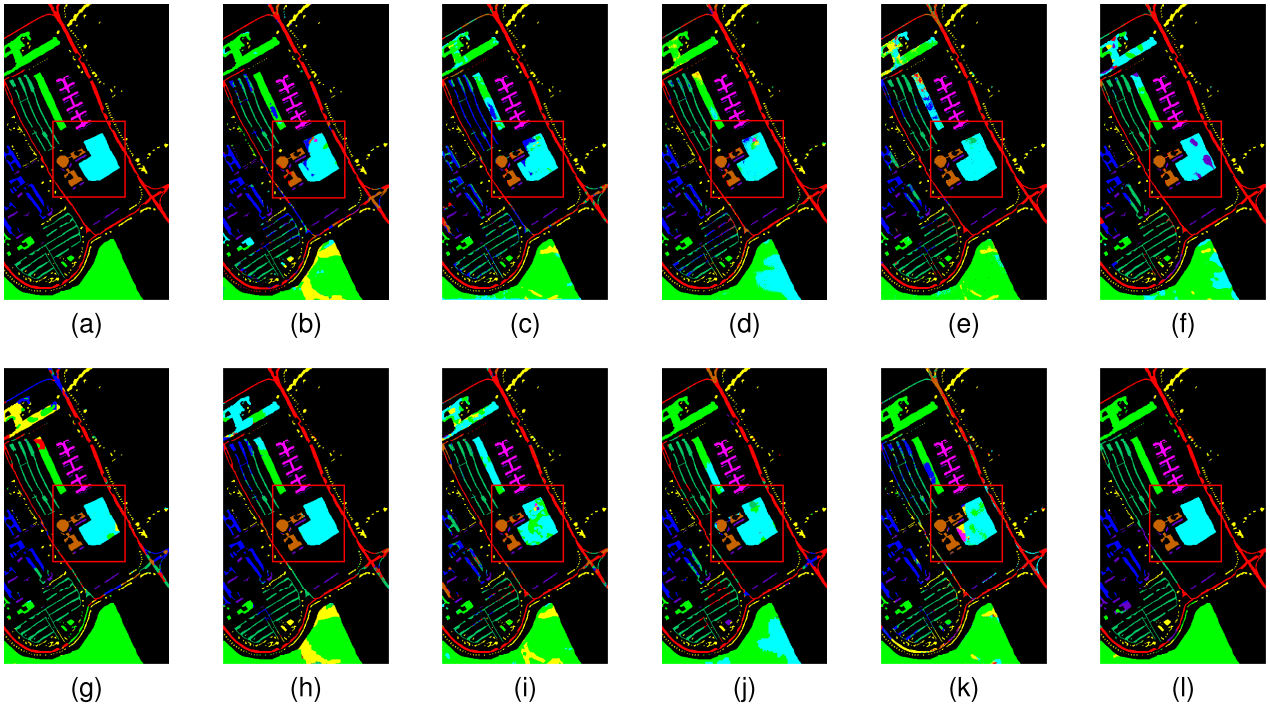


Fig. 6. Classification maps obtained by different methods in the UP dataset (five labeled samples per class for training). Red boxes highlight areas with significant performance differences. (a) Ground truth. (b) MambaHSI. (c) MLPA. (d) SSMAE. (e) SSEFN. (f) SPFormer. (g) AMGCFN. (h) CEGCN. (i) GAHT. (j) WSGraphCL. (k) Gia-CFSL. (l) DB-HGCN (ours).

reach 91.18% and 86.27%, with gains of 0.07%–9.38% and 2.58%–11.25%, respectively. It ranks first in 4 out of 9 categories, especially in Gravel and Bare Soil. Fig. 6 shows its superior spatial consistency.

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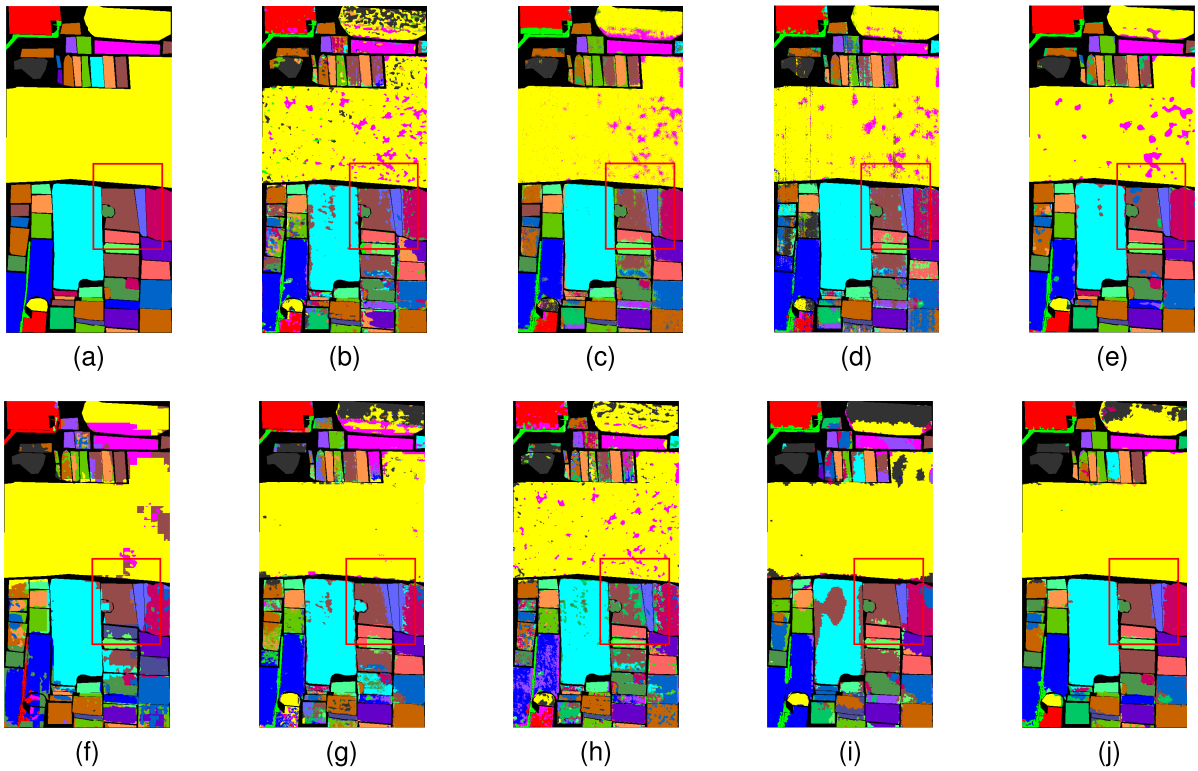


Fig. 7. Classification maps obtained by different methods in the HH dataset (five labeled samples per class for training). Red boxes highlight areas with significant performance differences. (a) Ground truth. (b) MambaHSI. (c) SSMAE. (d) SSEFN. (e) SPFormer. (f) AMGCFN. (g) CEGCN. (h) GAHT. (i) WSGraphCL. (j) DB-HGCN (ours).

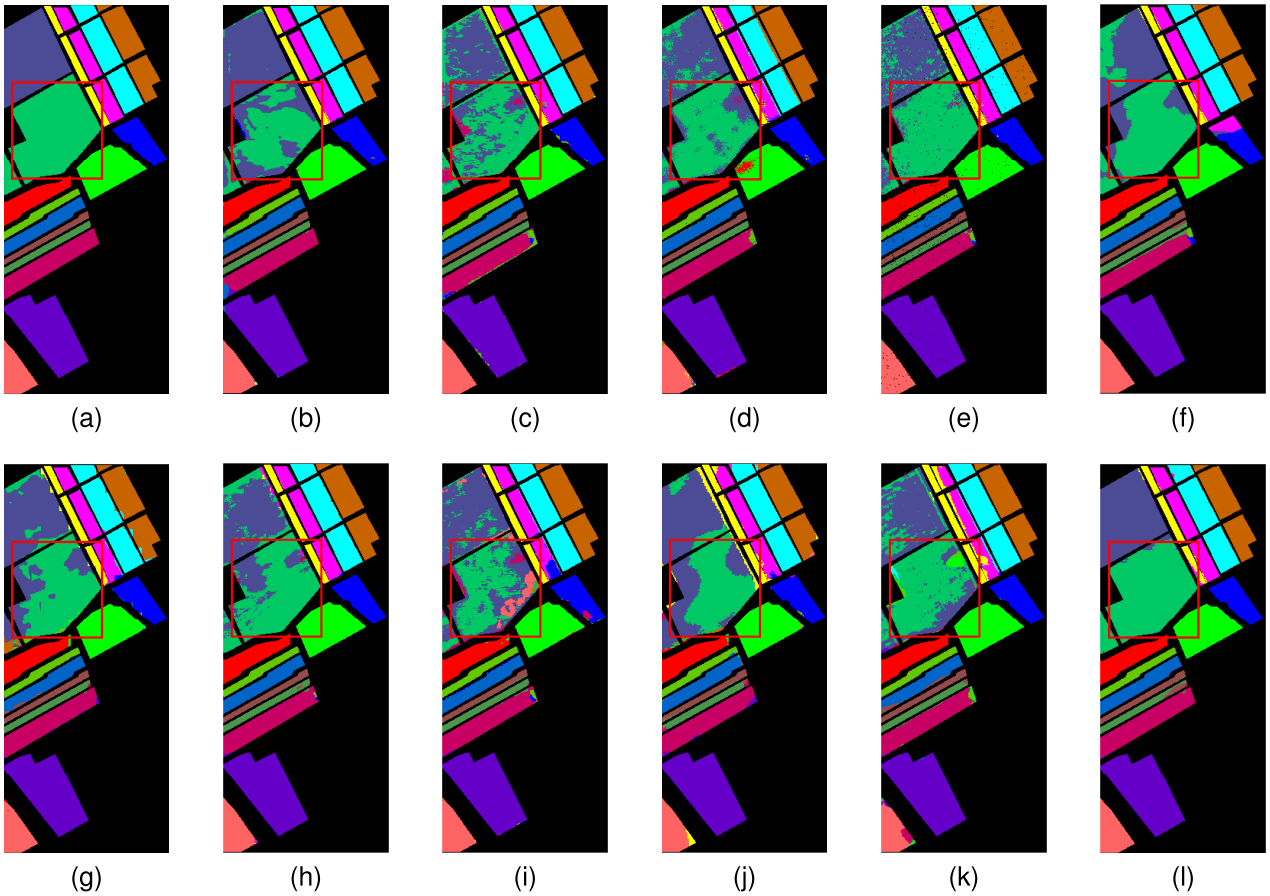


Fig. 8. Classification maps obtained by different methods in the SA dataset (five labeled samples per class for training). Red boxes highlight areas with significant performance differences. (a) Ground truth. (b) MambaHSI. (c) MLPA. (d) SSMAE. (e) SSEFN. (f) SPFormer. (g) AMGCFN. (h) CEGCN. (i) GAHT. (j) WSGraphCL. (k) Gia-CFSL. (l) DB-HGCN (ours).

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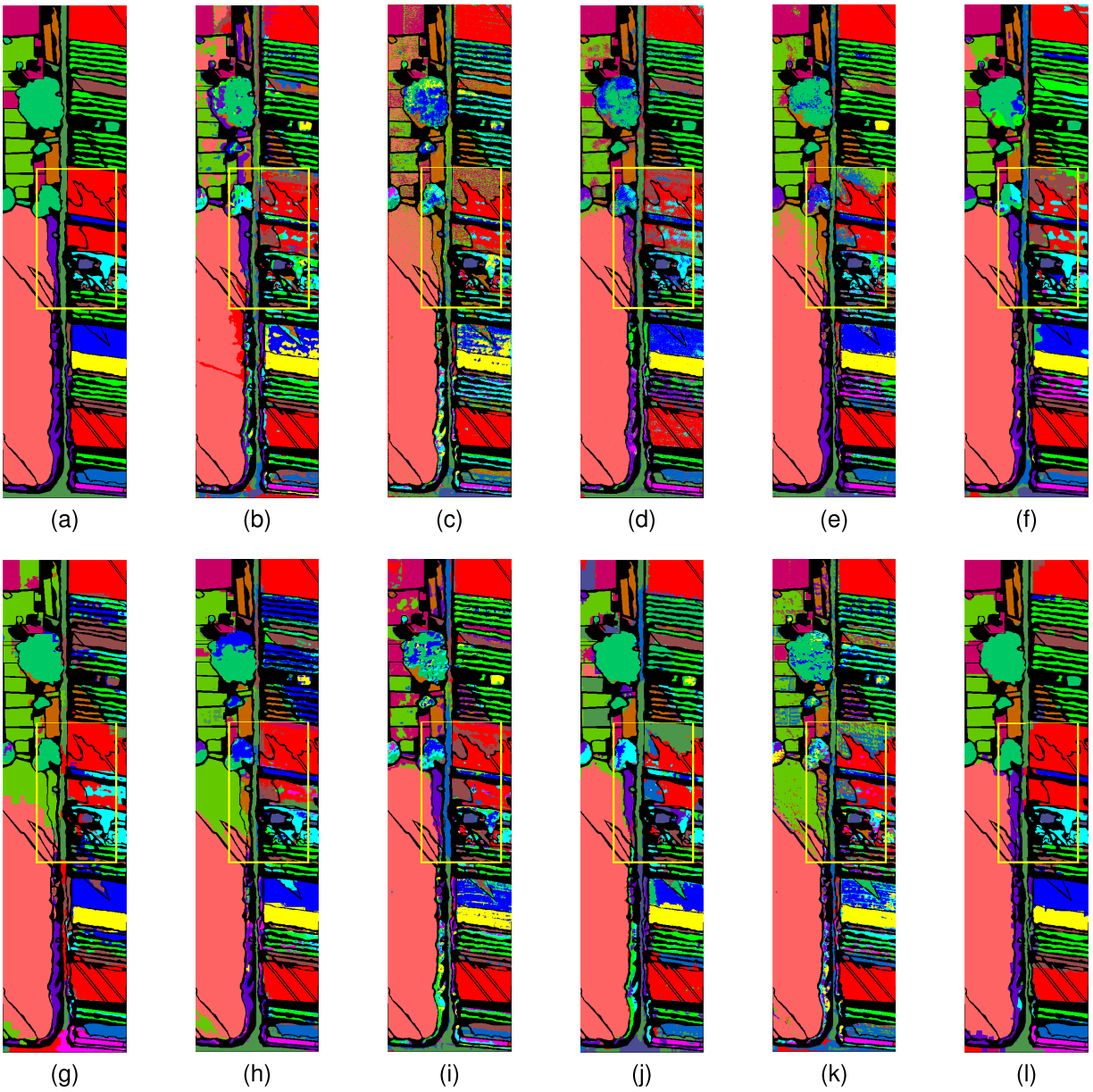


Fig. 9. Classification maps obtained by different methods in the HC dataset (five labeled samples per class for training). Yellow boxes highlight areas with significant performance differences. (a) Ground truth. (b) MambaHSI. (c) MLPA. (d) SSMAE. (e) SSEFN. (f) SPFormer. (g) AMGCFN. (h) CEGCN. (i) GAHT. (j) WSGraphCL. (k) Gia-CFSL. (l) DB-HGCN (ours).

2) HH Dataset: On the HH dataset, DB-HGCN achieves 87.47% OA, 88.27% AA, and 84.54% Kpp, outperforming others by 1.3%–9.69%, 2.14%–18.35%, and 0.87%–11.64%, respectively. It achieves the highest accuracy in 8 of 22 categories, particularly in the cotton firewood field and pakchoi field. As shown in Fig. 7, it offers better feature discrimination in complex regions.

3) SA Dataset: On the SA dataset, DB-HGCN reaches 97.69% OA, with improvements of 5.48% to 11.54%. The AA and Kpp are 97.99% and 97.42%, respectively, exceeding others by 1.8%–6.1% and 2.55%–12.78%. It leads in 8 of 16 categories, notably in the Vineyard series. Fig. 8 illustrates better classification uniformity.

4) HC Dataset: On HC, DB-HGCN achieves 82.15% OA, 83.06% AA, and 79.50% Kpp, outperforming others by 1.79%–14.98%, 1.57%–22.56%, and 0.44%–17.29%, respectively. It ranks first in 4 of 16 categories, especially in the soybean field and trees. Fig. 9 demonstrates its strengths in boundary delineation and regional consistency.

The comparison of the classification results across the four datasets clearly demonstrates the superior classification capabilities of DB-HGCN in limited sample scenarios.

E. Feature Fusion Effect Analysis

The DB-HGCN model consists of two branches that extract spatial–semantic and spatial–spectral features. To demonstrate the feature fusion effect of these two branches, we present qualitative classification maps and quantitative OA, AA, and kappa performance from each branch as well as the fused dual-branches. The classification results on the four datasets are shown in Figs. 10–13. As observed from these figures, the classification maps generated by SSeHGCN alone exhibit

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TABLE VI
OAs (%) OF ABLATION OF EACH MODULE IN DB-HGCN ON FOUR DATASETS

	Module	CLIP	PCA	HGCN1	SSRFM	HGCN2	UP	HH	SA	HC
(1)	**CLIP only**	✓	-	-	-	-	31.79	14.80	18.71	30.66
(2)	**CLIP+HGCN1**	✓	-	✓	-	-	79.72 ± 3.70	73.58 ± 1.79	89.84 ± 0.98	74.48 ± 4.27
(3)	**PCA+HGCN1**	-	✓	✓	-	-	71.26 ± 4.02	78.95 ± 2.87	92.81 ± 1.71	69.31 ± 4.62
(4)	**CLIP+PCA+HGCN1(SSeHGCN)**	✓	✓	✓	-	-	85.71 ± 4.35	83.83 ± 2.49	94.23 ± 1.14	78.73 ± 4.06
(5)	**HGCN2**	-	-	-	-	✓	81.94 ± 6.86	83.55 ± 3.51	95.55 ± 2.14	74.08 ± 5.08
(6)	**SSRFM+HGCN2(SSpHGCN)**	-	-	-	✓	✓	82.51 ± 4.69	84.11 ± 2.17	96.15 ± 1.73	75.96 ± 4.41
(7)	**PCA+HGCN1+SSRFM+HGCN2**	-	✓	✓	✓	✓	82.22 ± 5.07	82.99 ± 2.28	95.50 ± 1.67	76.45 ± 4.63
(8)	**CLIP+PCA+HGCN1+HGCN2**	✓	✓	✓	-	✓	87.71 ± 4.05	86.35 ± 2.00	96.62 ± 1.20	81.55 ± 4.38
(9)	**CLIP+PCA+HGCN1+SSRFM+HGCN2(ours)**	✓	✓	✓	✓	✓	$\mathbf{89.35 \pm 2.98}$	$\mathbf{87.47 \pm 1.63}$	$\mathbf{97.69 \pm 0.72}$	$\mathbf{82.15 \pm 4.44}$

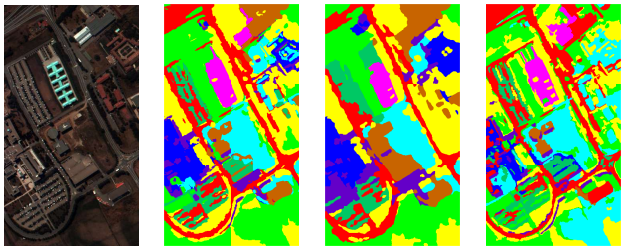


Fig. 10. Classification maps on the UP dataset. (a) False-color image. (b) DB-HGCN (OA = 89.51%, AA = 89.16%, and Kpp = 86.43%). (c) SSeHGCN Branch (OA = 86.73%, AA = 86.30%, and Kpp = 82.88%). (d) SSpHGCN Branch (OA = 74.31%, AA = 84.52%, and Kpp = 68.61%).

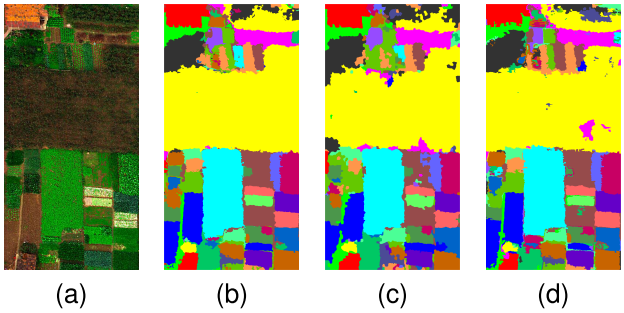


Fig. 11. Classification maps on the HH dataset. (a) False-color image. (b) DB-HGCN (OA = 92.14%, AA = 92.67%, and Kpp = 90.20%). (c) SSeHGCN Branch (OA = 84.04%, AA = 87.58%, and Kpp = 80.48%). (d) SSpHGCN Branch (OA = 87.52%, AA = 89.24%, and Kpp = 84.77%).

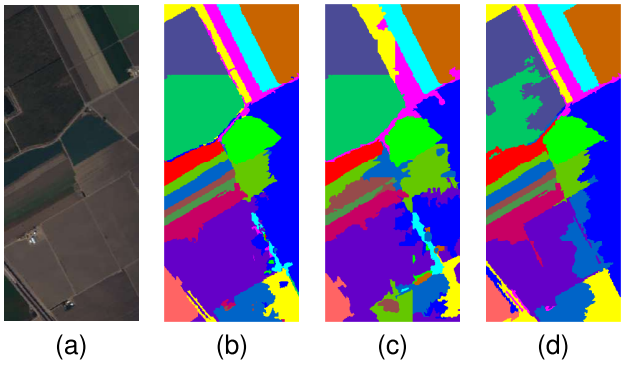


Fig. 12. Classification maps on the SA dataset. (a) False-color image. (b) DB-HGCN (OA = 97.66%, AA = 98.78%, and Kpp = 97.39%). (c) SSeHGCN Branch (OA = 90.40%, AA = 86.05%, and Kpp = 89.32%). (d) SSpHGCN Branch (OA = 91.78%, AA = 96.85%, and Kpp = 90.90%).

smoothness in large homogeneous regions but suffer from misclassifications in finer regions. In contrast, SSpHGCN delivers reliable classification in detailed areas but introduces

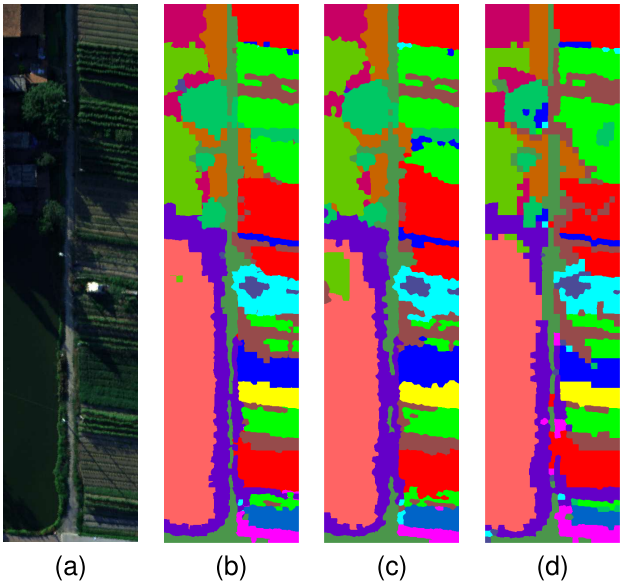


Fig. 13. Classification maps on the HC dataset. (a) False-color image. (b) DB-HGCN (OA = 88.19%, AA = 90.48%, and Kpp = 86.38%). (c) SSeHGCN Branch (OA = 86.04%, AA = 90.06%, and Kpp = 83.98%). (d) SSpHGCN Branch (OA = 83.18%, AA = 84.16%, and Kpp = 80.62%).

noise in more homogeneous regions. The fusion of these two branches, as illustrated in Figs. 10(b), 11(b), 12(b), and 13(b), results in the classification maps more like the ground truth. This fusion enhances edge delineation, maintains uniformity in large regions, and ensures accurate classification in fine-grained areas. The fused dual-branch model achieves significantly better OA, AA, and kappa than each individual branch, further demonstrating the effectiveness of feature fusion.

F. Ablation Analysis

The DB-HGCN model primarily relies on two branches for the extraction and fusion of distinct features, aiming to achieve enhanced classification accuracy. To assess the effectiveness of various modules, we designed ablation experiments, as shown in Table VI and Fig. 14. Additionally, to evaluate the role of hypergraph modeling under multihop constraints, we conducted constraint ablation experiments, as illustrated in Table VII and Fig. 15. The following analyses are based on the findings from these experiments.

1) Effectiveness of Clip-Based Semantic Module and PCA Feature Aggregation: To evaluate the impact of CLIP and

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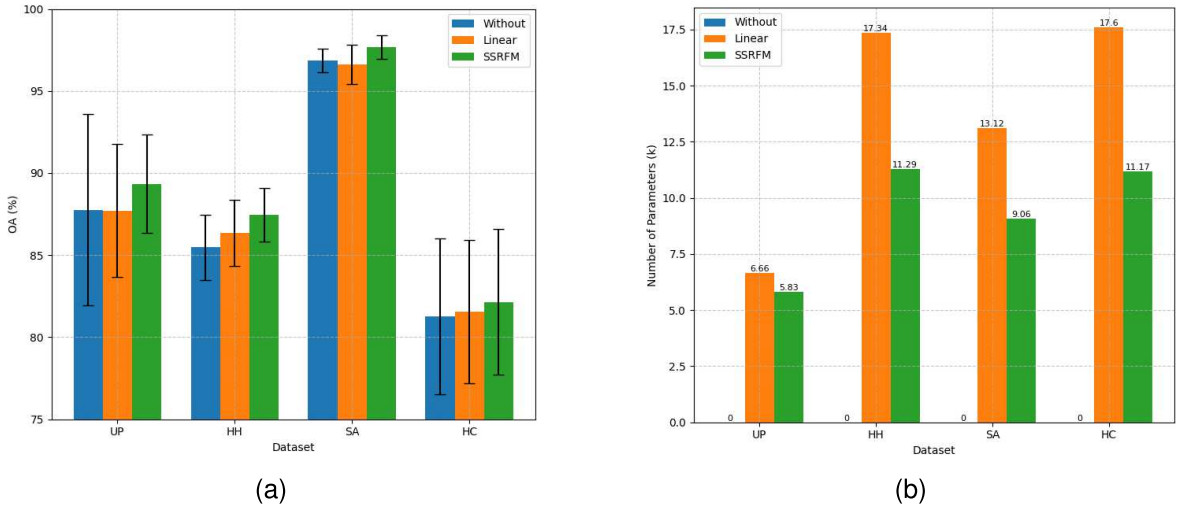


Fig. 14. Ablation analysis of SSRFM. (a) OAs (%) of different processing ways. (b) Number of parameters (k).

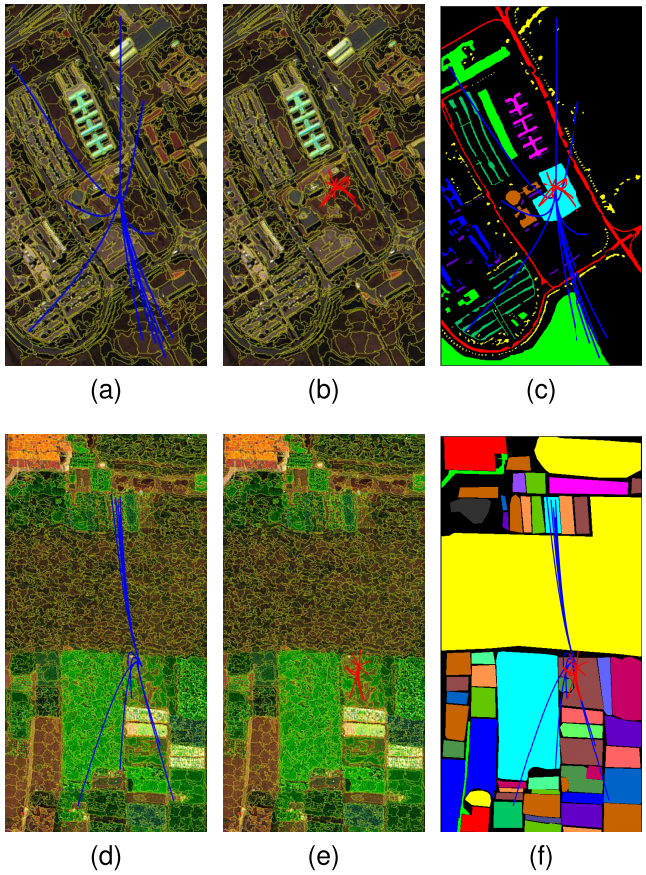


Fig. 15. Visualization of ablation analysis under multihop constraint. (a) and (b) Results on UP without/with constraint. (c) Comparison of unconstrained (blue) and constrained (red) hyperedges associated with the same superpixel on UP, overlaid on the ground truth. (d) and (e) Results on HH without/with constraint. (f) Comparison of unconstrained (blue) and constrained (red) hyperedges on HH, overlaid on the ground truth.

PCA feature aggregation, we conducted three experiments: using only PCA features, only CLIP features, and a combined approach that integrates both, as shown in Table VI (2)–(4). The experimental results across the four datasets indicate that the OA of (4), which integrates both CLIP and PCA features, increased by 5.99% and 14.45% on the UP dataset, 10.25% and 4.88% on the HH dataset, 4.39% and 1.42% on the SA dataset, and 4.25% and 9.42% on the HC dataset, compared to (2) and (3), respectively. These findings suggest that combining CLIP semantic features with PCA-based features enhances their individual effectiveness, leading to a significant improvement in OA. This result supports the proposed spatial–semantic enhancement feature construction method.

TABLE VII
OAs (%) OF ABLATION OF MULTIHOP CONSTRAINT ON FOUR DATASETS

Multi-hop Constraint	UP	HH	SA	HC
without	81.37 ± 5.76	81.75 ± 4.14	94.31 ± 1.62	79.93 ± 2.11
with	$\mathbf{89.35 \pm 2.98}$	$\mathbf{87.47 \pm 1.63}$	$\mathbf{97.69 \pm 0.72}$	$\mathbf{82.15 \pm 4.44}$

To further validate the CLIP-based semantic module, we conducted a comparison using experiment (7), where the CLIP-derived semantic prior is excluded while all other components remain. Despite leveraging a strong spatial–spectral backbone, experiment (7) shows inferior performance compared to the full model (9), with performance gains of 7.13%, 4.48%, 2.19%, and 5.70% on UP, HH, SA, and HC, respectively, when CLIP is included. Additionally, a zero-shot configuration (1), using only the pretrained CLIP model, shows limited performance due to the lack of sample-specific learning. Notably, on datasets where pretrained CLIP performs well (UP and HC with OA >30%), the semantic-enhanced hypergraph branch (SSeHGCN, 4) outperforms the spatial–spectral branch (SSpHGCN, 6), emphasizing CLIP's pivotal role in feature fusion and model performance.

2) Effectiveness of SSRFM: To evaluate the impact of SSRFM, we compared it with traditional linear layers used for feature dimensionality reduction within SSpHGCN. In experiment (5), linear layers were used, while (6) incorporated the SSRFM module. The results indicate that SSRFM improved OA by 0.57% on the UP dataset, 0.56% on the HH dataset, 0.6% on the SA dataset, and 1.88% on the HC dataset, compared to the linear layer approach. This confirms that the SSRFM module effectively enhances the representational

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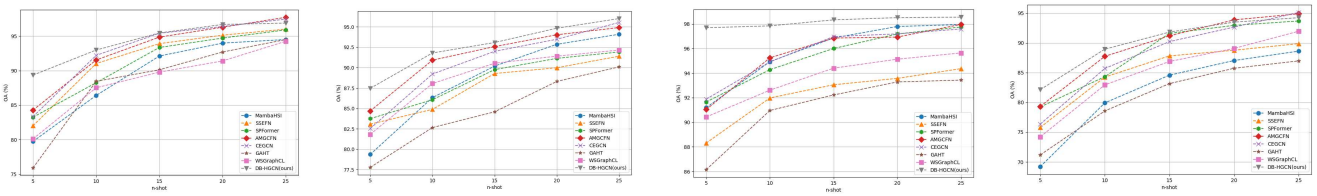


Fig. 16. Evolution of overall accuracy (OA) as a function of training samples per class. (a) UP. (b) HH. (c) SA. (d) WHU-Hi-HanChuan.

TABLE VIII
RUNNING TIME (S), PARAMETER SIZE (K), AND FLOPS (G) OF DIFFERENT METHODS

Dataset	Metric	MambaHSI	MLPA	SSMAE	SSEFN	SPFormer	AMGCFN	CEGCN	GAHT	WSGraphCL	Gia-CFSL	DB-HGCN(ours)
UP	**Preprocessing time(s)**	0.12	14.29	11901.44	1.03	144.51	6.81	5.76	0.09	189.28	12.24	16.89
	Training time(s)	134.36	620.20	80.51	26.60	2.24	52.31	28.07	6.14	4287.20	2887.62	9.08
	Testing time(s)	1.14	2.32	76.64	0.02	2.96	0.04	1.90	10.60		5.15	0.005
	Parameters(k)	412.24	4259.29	1356.29	12841.16	45.23	343.61	152.86	927.10	9959.04	428.19	102.97
	FLOPs(G)	39.18	226.55	27.48	161.48	0.49	29.50	26.11	8.15	-	8.94	1.95
HH	**Preprocessing time(s)**	1.03	-	19523.63	4.27	186.55	8.93	9.63	0.90	143.99	-	86.87
	Training time(s)	276.63	-	257.25	72.11	6.64	105.40	88.59	14.88	12604.80	-	40.26
	Testing time(s)	2.44	-	958.12	0.02	41.06	0.02	0.01	41.72		-	0.003
	Parameters(k)	435.29	-	1360.61	18893.26	89.34	386.70	176.24	1514.97	9730.05	-	116.09
	FLOPs(G)	103.49	-	34.28	237.12	2.35	102.86	77.38	32.55	-	-	10.26
SA	**Preprocessing time(s)**	0.28	37.08	8206.11	1.33	47.85	7.53	5.03	0.14	391.83	38.90	10.72
	Training time(s)	75.73	1331.61	577.25	51.86	4.79	33.13	21.36	10.72	6578.90	4610.98	4.88
	Testing time(s)	0.58	4.66	94.99	0.02	5.56	0.01	1.04	7.17		6.39	0.004
	Parameters(k)	426.07	4270.52	5384.06	16496.98	86.47	369.01	166.89	972.62	13862.91	439.30	110.32
	FLOPs(G)	23.86	402.75	194.84	207.23	2.32	21.58	17.15	15.18	-	27.84	1.84
HC	**Preprocessing time(s)**	0.82	294.71	15488.89	3.71	170.74	7.99	7.16	0.54	213.00	301.06	44.60
	Training time(s)	237.99	3487.28	154.01	52.55	5.71	86.59	56.10	13.86	11811.20	3931.21	18.77
	Testing time(s)	2.03	24.50	489.74	0.02	26.74	0.02	7.42	38.10		25.75	0.006
	Parameters(k)	435.03	4277.52	1360.61	19005.78	89.26	382.87	175.99	1519.18	9734.15	477.19	112.43
	FLOPs(G)	85.77	413.75	24.93	238.93	2.35	82.14	63.73	23.74	-	28.16	6.13

capacity of the SSpHGCN branch, resulting in improved performance.

To better illustrate SSRFM's performance, an ablation experiment was conducted within the dual-branch architecture. In this experiment, we compared linear and SSRFM as feature enhancement modules across the four datasets, with and without processing. As shown in Fig. 14(a), the results indicate that while the linear method marginally improves stability, its impact is limited. Notably, on the UP and SA datasets, the OA achieved using the linear method was lower than the baseline without processing. In contrast, the SSRFM method effectively mitigated fluctuations and improved accuracy. The results shown in Table VI (8) and (9) indicate that SSRFM enhanced OA by 1.64%, 1.12%, 1.07%, and 0.60% on the UP, HH, SA, and HC datasets, respectively. Additionally, Fig. 14(b) illustrates the parameter consumption for the methods, showing that SSRFM requires fewer parameters than the linear method. This demonstrates that SSRFM can achieve a more effective performance on feature enhancement and denoising with fewer parameters.

3) Effectiveness of the Multihop Constraint Hypergraph Construction Method: To evaluate the effectiveness of hypergraph construction under multihop constraints, we conducted experiments comparing the method with and without multihop constraints, as shown in Table VII. The experimental results across the four datasets clearly show that hypergraph modeling with multihop constraints outperforms methods without such constraints. Compared to the approach without multihop constraints, the best OA under the multihop constrained method increased by 7.98%, 5.72%, 3.38%, and 2.22% on the UP, HH, SA, and HC datasets, respectively.

We further visualized the impact of multihop constraints on the UP and HH datasets, as illustrated in Fig. 15. Specifically, for a selected superpixel node, we visualized all the hyperedges it participates in, comparing the resulting hypergraph structures with and without multihop constraints. The visualizations in subfigures (a), (b), (d), and (e) show the overall structural differences, while (c) and (f) provide a more detailed comparison by overlaying the corresponding hyperedges on the ground-truth map. The results clearly demonstrate that introducing multihop constraints significantly reduces noisy or irrelevant connections in the hypergraph. Even in complex or highly heterogeneous regions, the constrained hypergraph structure maintains better consistency and stability.

In conclusion, the use of multihop constraints during hyperedge construction effectively introduces structural regularization and guides the modeling of structural priors in the hypergraph.

G. Generalization Performance

To evaluate the generalization performance of DB-HGCN, we conducted experiments with sample sizes of [5, 10, 15, 20, 25] and compared the results with other models, as shown in Fig. 16. The results reveal that DB-HGCN outperforms other models in most cases, especially with limited sample sizes. However, as the sample size increases, the advantage of the semantic features from the pretrained model diminishes, preventing DB-HGCN from achieving the highest accuracy. Nonetheless, DB-HGCN consistently remains among the top-performing models, demonstrating its strong generalization capability.

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H. Complexity Analysis

To assess the computational efficiency of our model, we compare DB-HGCN with ten representative methods in terms of preprocessing/training/testing time, parameter count, and FLOPs, as shown in Table VIII. DB-HGCN only consists of two hypergraph convolutional blocks and several linear projections, all implemented via matrix multiplication. Although it incorporates SLIC and a pretrained semantic module, these preprocessing steps are performed only once and shared throughout the training process. Once completed, both training and inference proceed efficiently. In terms of training speed, DB-HGCN is the fastest among all non-Transformer models. The testing phase is highly efficient, with inference times consistently below 0.01 s across all datasets.

DB-HGCN has achieved the smallest or near-smallest parameter count across all four datasets, along with significantly reduced FLOPs. Additionally, the average runtime memory usage is 8.9 GB, which is well within the capacity of standard GPUs. These results suggest that DB-HGCN maintains a good balance between efficiency and performance, making it a competitive and scalable choice for real-world deployment.

IV. Conclusion

In this article, we propose a DB-HGCN for HSI classification in scenarios with limited samples. The DBHGCN architecture consists of two branches: SSeHGCN and SSpHGCN. The SSeHGCN branch extracts semantic features from HSI using CLIP, which enhances the spatial representation of primary spectral information extracted via PCA. The SSpHGCN branch utilizes the full spatial–spectral information to analyze and represent the HSI, especially with the SSRFM module effectively fuses visual features within superpixels and the relational features between them. Both branches capture high-order associations among superpixels through hypergraph convolution, and the multihop constraint in the hypergraph modeling guides the structural prior modeling of the HSI. Experimental results on four datasets show that DB-HGCN demonstrates exceptional feature extraction abilities for HSI classification with limited samples, significantly improving classification accuracy.

In conclusion, this study presents a novel approach for HSI classification under limited sample conditions, integrating semantic knowledge with spatial–spectral analysis via superpixel-based hypergraph modeling. It highlights the potential of spatial–spectral–semantic feature integration and the application of hypergraph theory to enhance HSI data processing. Future work will focus on further refining and exploring the integration of prior knowledge, including exploration of more advanced semantic fusion methods beyond CLIP.

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